

ISO26262 Functional Safety Report

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Development Method: **ATDD**

Field	Value
Document Template	safety_report
Development Method	ATDD
Requirement	safe monitoring of pedestrian on sidewalk in bus-stop area
Related ISO Parts	Part 11 - Semiconductors Part 4 - System Development Part 9 - ASIL-Oriented Analysis
Referenced Pages	140, 16, 163, 177, 178, 179, 9

ISO26262 Evidence

Part: Part 11 - Semiconductors

Clause: Unknown

Page: 9

Score: 0.4491

Summary: ■ Road vehicles — Functional safety — Part 11: Guidelines on application of ISO 26262 to semiconductors 1 Scope This document is intended to be applied to safety-related systems that include one or more electrical and/or electronic (E/E) systems and that are installed in series production road vehicles, excluding mopeds. This document does not address unique E/E systems in special vehicles such as E/E systems designed for drivers with disabilities. NOTE Other dedicated application-specific safety standards exist and can complement the ISO 26262 series of standards or vice versa. Systems and their components released for production, or systems and their components already under development prior to the publication date of this document, are exempted from the scope of this edition. This docu

Part: Part 11 - Semiconductors

Clause: 1.1

Page: 140

Score: 0.4399

Summary: ■ ISO 26262-11:2018(E) Annex A (informative) Example on how to use digital failure modes for diagnostic coverage evaluation A.1 Example of evaluation of a DMA safety mechanism A.1.1 Description of the use case The following is the DMA use case considered in this example: ■ a message is received by a communication peripheral every X ms; ■ as soon as the message is received by the communication peripheral, it triggers a DMA request; ■ the DMA transfers the message from the peripheral receive buffer to a RAM region; ■ the transfer is always to the same RAM region, independent from the message content; ■ after the DMA is finished with the transfer, it triggers a CPU interrupt; and ■ the CPU copies the message into a different buffer within the RAM depending on the message ID. A.1.2 Description

Part: Part 9 - ASIL-Oriented Analysis

Clause: Unknown

Page: 178

Score: 0.4157

Summary: ■ ISO 26262-11:2018(E) The derived safety requirement for the PLD could be: ■ SafReq_PLD_001: "Output of a value larger than the minimum of the two input values from µController1 and µController2 shall be avoided" (derived from SafReq_hardware_Comp_Controller_001); and ■ SafReq_PLD_002: "Undetected corruption of the CAN output value from PLD which leads to an output too high shall be avoided" (derived from SafReq_hardware_Comp_Controller_001). The following clause addresses, as an example, two different approaches for the PLD's safety and dependent failure analysis. The safety analysis and the dependent failure analysis concerning µController1 and µController2 are out of scope of this document. Failures of the PLD can be addressed by two approaches: — utilizing safety measures which are e

Part: Part 11 - Semiconductors

Clause: 5.2.3

Page: 163

Score: 0.4119

Summary: ■ ISO 26262-11:2018(E) Annex D (informative) Examples of quantitative analysis for analogue component D.1 Description The following is an example of a quantitative analysis using the method described in 5.2.3 in order to calculate the single-point fault metric and the latent-fault metric for a given safety requirement allocated to the mixed signal hardware element depicted in Figure D.1. The example consists of a mixed signal hardware element composed of: ■ a low drop voltage regulator (low drop voltage regulator in Figure D.2) providing an output voltage within a prescribed range; ■ a voltage monitor (voltage monitor in Figure D.3) capable of detecting overvoltage ($V_A > OV_{th}$) and under-voltage ($V_A < UV_{th}$) on the LDO output by monitoring the regulated voltage V_A and comparing it with two p

Part: Part 9 - ASIL-Oriented Analysis

Clause: Unknown

Page: 179

Score: 0.4048

Summary: ■ ISO 26262-11:2018(E) Table E.1 — Example of a PLD safety analysis in the case of PLD external measures Failure mode Permanent distribution Transient distribution PVSG? MPF? Safety mechanisms FM_PLD_OP_01: no output 14,3 % 14,3 % 1 0 SafMech_Network_001 FM_PLD_OP_02: output of old message 14,3 % 14,3 % 1 0 SafMech_PLD_001 FM_PLD_OP_03: corrupt output 14,3 % 14,3 % 1 0 SafMech_PLD_001 FM_PLD_OP_04: does not out- put minimum value 14,3 % 14,3 % 0 1 SafMech_PLD_001 FM_PLD_OP_05: always out- puts μ Controller1 value 14,3 % 14,3 % 0 1 FM_PLD_OP_06: always out- puts μ Controller2 value 14,3 % 14,3 % 0 1 SafMech_PLD_001 FM_PLD_OP_07: active “Disa- ble” discrete signal does not prevent CAN transmission 14,3 % 14,3 % 0 1 NOTE PVSG = potential to directly violate the safety goal; MPF = multiple-point

Part: Part 11 - Semiconductors

Clause: Unknown

Page: 177

Score: 0.4022

Summary: ■ ISO 26262-11:2018(E) Annex E (informative) Examples of quantitative analysis for PLD component E.1 Architecture of the example Figure E.1 shows the system used in the following examples. The system is intended for a safety- related application where two microcontrollers are used for redundancy and the final control output is implemented using a PLD. The two microcontrollers send their values to the PLD via SPI (Serial Peripheral Interface) and the PLD communicates via CAN (Controller Area Network) bus. For this example, it is assumed that a calculated output too high (i.e. greater than the value that would have been determined by a non-faulted system plus a threshold) is a potential hazard but an output too low is acceptable from a functional safety point-of-view. It is also assumed that

Part: Part 4 - System Development

Clause: Unknown

Page: 16

Score: 0.4018

Summary: 26262-2, ISO 26262-4, ISO 26262-5, ISO 26262-6 (in the case of software based safety mechanisms to cover hardware failures), ISO 26262-8, and ISO 26262-9, whenever applicable, can be used for the design of the safety mechanisms during the development of the IP. EXAMPLE 1 Bus “fabric” with built-in bus supervisors including fault detection and notification logic (e.g. interrupt signals). EXAMPLE 2 Voltage regulator with monitoring (under-voltage and over-voltage detection), protection (current limit or thermal protection) and self-diagnostics (monitoring and protection circuit built-in self-tests). Alternatively the IP can be developed with no assumed safety requirements or specific safety mechanisms to detect and control faults. EXAMPLE 3 Bus “fabric” without built-in bus supervisors or er

Hazard / ASIL

Pedestrian collision at bus stop due to failure in pedestrian monitoring system

ASIL: ■ ■ ■ ■

Safety Goal / FSR / TSR

Safety Goal:

■ ■ ■ ■

FSR:

■ ■ ■ ■

TSR:

■ ■ ■ ■

ATDD Acceptance Criteria

TC-001 - Pedestrian detection and stop

Given: Pedestrian enters bus stop area within 5 meters

When: System detects pedestrian

Then: Bus stops and alerts driver

Expected Action: Bus stops and triggers audible alert

Verification Method: Check bus stop signal and driver alert response

Methodology: ATDD

Traceability

Development Method: ATDD

Requirement: REQ-001

Hazard: HZ-001

Safety Goal: SG-001

FSR: FSR-001

TSR: TSR-001

Test Cases: TC-001

Trace Chain:

REQ-001 → HZ-001 → SG-001 → FSR-001 → TSR-001 → TC-001